Python Regression– Analysis Questions

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Analysis Questions

1. What does the sign of the slope tell you physically about the relationship between temperature and dissolved oxygen?

The slope being negative shows that there is a corresponding negative relationship between temperature and oxygen levels. What this means is that as temperature increases, the dissolved oxygen concentration decreases in turn.

2. Your R^2 value should be close to 1.0. What does this indicate about how well temperature explains dissolved oxygen variation?

The R^2 value is 0.996, showing that the linear regression almost perfectly describes the actual variation in the recorded data. What this means in concrete terms is that the relationship between temperature and oxygen concentration in this dataset is almost perfectly linear, and that temperature measurements very accurately predict dissolved oxygen levels.

3. Do the data points appear to follow a linear trend? Additionally, what might suggest that a nonlinear model would work better?

The data points do follow a downward sloping linear trend. Nothing from this dataset obviously indicates that any other model would more accurately describe the trend. If anything, the somewhat varied differences in the 0 to 100^{th} percentiles shows that the relationship might not be perfectly linear, especially outside the measured data range, nevertheless a linear model very accurately describes the data in this range.

4. Why would it be physically unreasonable to use this regression model to predict dissolved oxygen at 0 °C? 50 °C? What is this modeling limitation called?

It would be unreasonable to extrapolate it to 50°C, probably more so for 0°C because that is the point at which water freezes and other considerations come into play. Although the relationship presents itself to be pretty much linear from 14-28°C, so extending the measurements that far, over multiple ranges of the dataset would likely be an inaccurate extrapolation.

5. Site C is downstream and tends to have lower dissolved oxygen than Sites A and B. Suggest ONE engineering or environmental reason why downstream dissolved oxygen may be lower.

One reason that the dissolved oxygen decreases downstream could be that the water generally gets warmer more downstream. This could be due to the water getting more sunlight as it flows more which means that downstream water is overall warmer than midstream water which in turn is warmer than upstream water. This is also observed and supported by the data.